

# Practice Quiz: Systems (Implementation)

Name: \_\_ Date: \_\_

## Part A: Multiple Choice

1. In the Active Inference implementation, the A matrix has shape: A) (num\_states, num\_observations) B) (num\_observations, num\_states) — rows are observations, columns are states C) (num\_actions, num\_states) D) (1, num\_states)
2. The B matrix has 3 dimensions because: A) It represents 3D space B) It stores a separate state-to-state transition matrix for each action C) Python requires 3D arrays D) There are always exactly 3 states
3. The softmax function converts: A) Probabilities to log-probabilities B) Log-scores to normalized probabilities where higher scores get higher probability C) States to observations D) Integers to floats
4. Numerically stable log uses  $\log(x + \epsilon)$  because: A) It makes computations faster B) It prevents  $\log(0) = -\text{infinity}$  which would crash the program C) Python requires it D) It's more accurate
5. The normalize function ensures: A) All values are between -1 and 1 B) The array sums to 1, making it a valid probability distribution C) All values are positive integers D) The array is sorted
6. To validate a generative model, you should check that: A) All matrices contain positive integers B) Columns of A and B sum to 1, and D sums to 1 C) All values are greater than 0.5 D) The matrices are square
7. KL divergence  $D_{\text{KL}}(q \parallel p) = 0$  when: A) q and p are very different B) q and p are identical distributions C) q is uniform D) p is uniform

## Part B: Short Answer

1. Write Python code to create an A matrix for a 3-state, 2-observation system where observation is 85% reliable. Include the validation check.
2. Implement a Bayesian update function that takes A, prior, and observation\_index as inputs and returns the normalized posterior. Show a test case.
3. Explain why implementing Active Inference in code is valuable for understanding the theory. Give a specific example of something that becomes clearer through implementation.